UMKC Impact & Translation Series

Regulatory Considerations for University Startups

Helping you move your work beyond the university

Work that moves beyond the university and into real-world use often encounters regulatory requirements. In some cases, this includes forming a startup. In others, it involves partnerships, licensing, or adoption within existing systems.

Whether you're developing software, launching a medical device, or commercializing a new therapeutic, understanding the legal, institutional, and federal guidelines early can help you avoid delays and build a stronger path to market.

This whitepaper offers a clear, early look at regulatory issues that affect university startups—especially those in biotech, medical devices, and life sciences. Our goal is to help you understand what may come up, what to prepare for, and when to seek expert advice.

UITS Whitepaper

This whitepaper is part of the **UMKC Impact & Translation Series (UITS)** - a set of short videos and supporting materials designed to help faculty think more clearly about how their work can move beyond the university and into real-world use.

The videos introduce core ideas. This document goes deeper on one of those ideas: **how regulatory requirements shape the path from research to real-world use.**

We'll also dig deeper into the federal pathways for biomedical startups, what compliance really looks like in practice, and how your regulatory strategy can help you raise funds, partner with others, and go to market successfully.

Choosing a Legal Structure and Understanding University Policies

One of your earliest decisions will be selecting the right legal structure for your startup. This choice influences your liability, tax obligations, fundraising strategy, and even how you compensate your team.

Most university startups choose a:

- **Limited Liability Company (LLC)** for simplicity, pass-through taxation, and operational flexibility
- **C-Corporation** for scalability and to attract venture capital or equity investors (especially if planning to raise significant funds or offer stock options)

Beyond standard business considerations, university-affiliated startups must also follow institutional rules. At UMKC and other research universities, these include:

- **Conflict of Interest (COI)** policies: Financial interests must be disclosed, and faculty/staff must avoid situations where university roles conflict with company roles.
- **Use of Resources**: University labs, equipment, and student time can't be used for company work without proper agreements like Sponsored Research Agreements or Facilities Use Agreements.
- **Conflict of Commitment (COC)**: Faculty and researchers must balance university time with outside ventures and stay within effort limits.

If you're a student, postdoc, or faculty member considering launching a startup, contact your Office of Commercialization early. You'll want to be transparent and get help navigating these boundaries before conflicts arise.

Intellectual Property and Licensing

Chances are that your innovation is built on IP that originated in a university lab. This means the university likely owns the IP, and your startup will need to license it for commercial use.

At UMKC, intellectual property protection and commercialization are handled by separate but coordinated functions. The Office of Technology Transfer (OTT) manages invention disclosure, patenting, and licensing. The Office of Commercialization (OC) focuses on helping you understand how your work could move into use, including startup pathways, partnerships, and regulatory positioning.

Understanding the terms of this license is crucial to your success. Here are a few concepts to keep in mind:

- **University Ownership**: Most universities automatically own inventions made using significant university resources or created within the scope of employment.
- **Exclusive vs. Non-exclusive Licensing**: Exclusive licenses give your startup a competitive advantage but often include royalty and diligence terms. Non-exclusive licenses may be more affordable but open to competitors.
- **Equity and Royalties**: Licensing deals typically involve upfront fees, milestone payments, royalties on sales, and sometimes university equity in your company.

It's a good idea to discuss expectations early. Ask about:

- Royalty rates
- Minimum performance milestones
- Rights to sublicense
- Fields of use and territory

The more you understand your licensing agreement, the better positioned you are to negotiate investor terms and protect your startup's value.

FDA Pathways for Biotech, Drug, and Device Startups

If you're developing a product that touches health—from wearable sensors to therapeutics—you'll encounter the FDA. Knowing what to expect from a regulatory standpoint is essential for designing your timeline, budget, and milestones.

Let's explore the core regulatory pathways for life sciences startups:

Drug and Biologic Development

When your startup is working on a new drug, vaccine, or biologic therapy, it enters one of the most highly regulated spaces in all of entrepreneurship. Unlike software or general consumer products, these innovations affect human health directly and are therefore subject to extensive scrutiny by the Food and Drug Administration (FDA).

Understanding the steps involved in drug and biologic development can help you design your R&D activities more effectively and align your funding strategy with regulatory milestones.

Drugs and biologics must be shown to be safe and effective before they can be sold. To begin testing in humans, companies must file an **Investigational New Drug (IND)** application with the FDA.

The IND must include:

- Preclinical animal study data conducted under **Good Laboratory Practices (GLP)**
- Chemistry, manufacturing, and control (CMC) information
- Clinical protocols and investigator background

Depending on the product type:

- Traditional drugs file a **New Drug Application (NDA)** with the Center for Drug Evaluation and Research (CDER)
- Biologics file a **Biologics License Application (BLA)** with the Center for Biologics Evaluation and Research (CBER)

Biologics include things like vaccines, cell and gene therapies, and monoclonal antibodies—each with its own complexity and regulatory nuance.

Medical Devices

Medical devices present a wide spectrum of regulatory complexity, ranging from simple consumer tools to high-risk, life-sustaining technologies. Understanding where your innovation fits into the FDA's classification system is critical not only for compliance but also for determining the development timeline, budget, and commercialization strategy.

Unlike drugs and biologics, devices are categorized by risk, which influences whether your product goes through a streamlined clearance process or a rigorous approval path. This section outlines the core device classifications and the regulatory routes associated with each.

Devices are classified into three regulatory classes depending on their risk:

- **Class I**: Low risk, often exempt from premarket review (e.g., tongue depressors)
- **Class II**: Moderate risk; usually requires a **510(k)** submission showing "substantial equivalence" to an existing approved device
- **Class III**: High risk; needs a full **Pre-Market Approval (PMA)**, including clinical data and rigorous review

Key points for startups:

- **510(k)** clearance is faster and less expensive than PMA, but requires a clear predicate device
- **Design controls** and **validation protocols** are necessary from early stages
- **Human Factors testing**, labeling reviews, and software validation can all trigger additional requirements

If you're unsure about your device classification, the FDA offers Pre-Submission (Q-Sub) meetings to clarify expectations.

Regulation Beyond Life Sciences

While regulatory pathways are most visible in biotech and medical devices, other fields also encounter forms of regulation.

Examples include:

- Data privacy and security requirements in software and AI
- Environmental and safety standards in engineering and materials
- Professional or accreditation standards in education and practice-based fields

The level of regulation varies, but the underlying principle is the same: real-world use often requires meeting external standards beyond academic validation.

Quality Assurance and Documentation

In regulated industries, how you develop your product can be just as important as the product itself. Quality assurance (QA) and quality control (QC) frameworks are designed to ensure safety, repeatability, and compliance throughout the product lifecycle.

Startups in the early stages may not need a full-fledged quality management system, but they do need to understand the fundamentals. Establishing good habits early—such as tracking experiments, validating designs, and documenting protocols—can prevent costly setbacks later.

Quality isn't just for big companies—founders must start building documentation habits early. Regulators will want to see traceability, validation, and a commitment to product safety.

You don’t need a full GMP system on day one, but you should:

- Maintain lab notebooks and SOPs (standard operating procedures)
- Document design decisions and risk analyses
- Plan for ICH Q10 (pharma) or 21 CFR Part 820 (devices) compliance over time

Outsourcing quality or regulatory tasks is common, but startups are still responsible for ensuring the work is done correctly. Auditable documentation will serve you well during fundraising, audits, and eventual scale-up.

Regulatory Strategy Shapes Your Business Plan

Too often, regulatory planning is treated as a technical detail that gets sorted out late in the game. But for university startups (especially those working in health, life sciences, or data-sensitive fields), your regulatory strategy directly influences nearly every other element of your venture. It determines your capital needs, hiring plan, product roadmap, and even the type of investors you'll attract.

Thinking strategically about regulation from the start allows you to avoid common pitfalls, answer investor questions with confidence, and maintain momentum as your technology matures.

Regulatory planning is not a back-office function; it's central to your startup's risk profile and go-to-market strategy.

Work that integrates regulatory thinking early is better positioned to move toward real-world use. For example, it helps to:

- Set realistic timelines and milestones
- Plan capital raises around major submissions (e.g., IND, 510(k), PMA)
- Align reimbursement strategy with product classification
- Attract investors with clearer exit potential

Ask yourself:

- How long will it take to get market clearance or approval?
- What does the total cost of compliance look like?
- Do we need a regulatory consultant or full-time head of quality?
- Can we design for a lower-risk category?

The earlier you know your pathway, the better you can tailor your development, hiring, and fundraising strategies.

NIH, SBIR/STTR, and Government Funding Rules

Many university startups pursue federal funding through Small Business Innovation Research (SBIR) and Small Business Technology Transfer (STTR) programs. These can be essential lifelines for early-stage R&D but come with compliance responsibilities.

Guiding Question:

What regulatory requirements could shape how your work moves from research into real-world use, and what would you need to prepare for early?

Here's what to consider:

- **Principal Investigator Eligibility**: STTRs require formal collaboration with a research institution. The PI can be from the university or the company, but their level of effort must be documented.
- **Financial Conflict of Interest (FCOI)**: The NIH requires annual disclosures, management plans, and transparency if an investigator holds equity or a leadership role in the company.
- **Effort Reporting**: You must clearly track how much time each funded individual is spending on the project, especially if they are university-affiliated.
- **Indirect Costs and Subawards**: These require prior negotiation with the cognizant federal agency. Many startups fail to budget correctly or mismanage subcontracts.

The NIH Grants Policy Statement (NIHGPS) provides detailed rules, but your university grants office can also assist.

Data Privacy and Security Regulations

Even non-healthcare startups today deal with sensitive personal data. Whether it's patient records, user behavior, or research results, startups must implement safeguards from the start.

Common regulations include:

- **HIPAA**: Applies to any business handling protected health information (PHI)
- **GDPR**: European Union regulation with strict rules on consent, storage, and international transfer
- **CCPA**: California-specific regulation focused on consumer privacy and opt-out rights

Best practices include:

- Using encrypted cloud storage
- Limiting access to sensitive data
- Maintaining data use agreements (DUAs) and privacy policies

If your business model involves AI or machine learning with personal data, pay extra attention. Transparency and explainability are becoming compliance requirements.

University and Community Resources

No founder is expected to be a regulatory expert on day one. That's why leveraging your university and regional innovation ecosystem is so important. These support systems exist to help you not only navigate compliance but also envision a larger path to impact for your innovation.

At UMKC, the **Office of Technology Transfer** is your first stop for questions about invention disclosure, IP ownership, and licensing agreements. They ensure that intellectual property is protected and university policies are followed.

The **Office of Commercialization** builds on that foundation. We help you discover and execute a plan to bring your innovation from lab to impact—through market exploration, startup support, regulatory education, and partner connections.

Helpful resources include:

- **Office of Technology Transfer**: Manages invention disclosure, IP strategy, and licensing
- **Office of Commercialization**: Supports startup development, commercialization strategy, and regulatory readiness
- **Research Compliance Office**: Helps you stay aligned with IRB, COI, and federal rules
- **Local accelerators and incubators**: Many have mentors or partner discounts on legal and regulatory consulting
- **SBDC or Biotech-focused mentoring groups**: These can assist with FDA navigation, pitch readiness, and funding strategy

At UMKC, we encourage startups to reach out early and often. Contact the **Office of Commercialization** at techcomm@umkc.edu or 816-235-6373.

Conclusion

Regulatory requirements are often seen as constraints. In practice, they are part of the structure that allows work to move safely and credibly into real-world use.

Understanding these requirements early does not slow progress. It clarifies what progress actually requires.

As your work begins to move beyond the university, regulatory considerations will influence what is possible, how long it will take, and what resources are needed.

You do not need to have all the answers. But recognizing that these factors exist and engaging early to understand them can significantly improve your ability to move your work toward meaningful impact.

Resources

UMKC Office of Commercialization

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UMKC Office of Technology Transfer

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📍 Website: https://ori.umkc.edu/facilities-compliance-and-commercialization/commercialization/process/submit-an-invention-disclosure.html

- FDA: Center for Drug Evaluation and Research (CDER)
- FDA: Center for Biologics Evaluation and Research (CBER)
- FDA: Center for Devices and Radiological Health (CDRH)
- NIH Grants Policy Statement (NIHGPS)
- UMKC Research Compliance Office